

Mechanical Ventilation Basics



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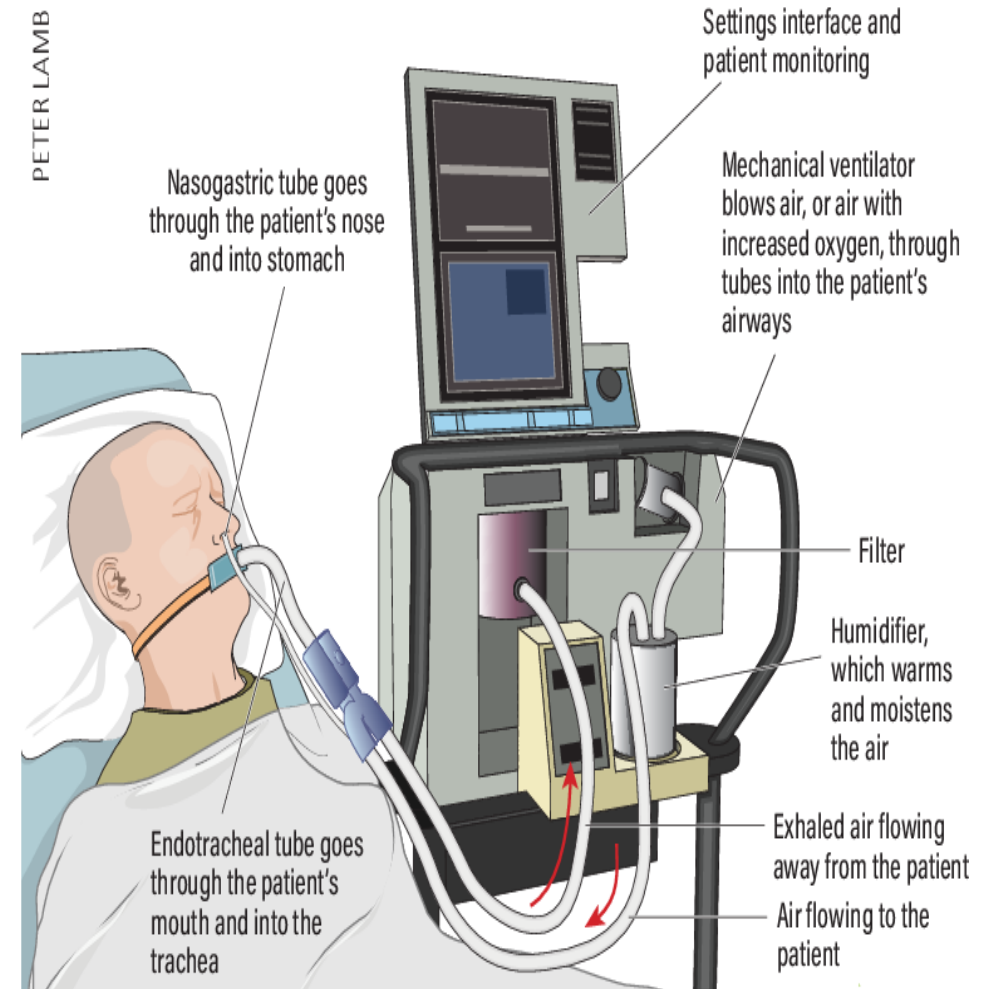
Outlines

- Introduction
- Definition
- Indications of Mechanical Ventilation
- Common ventilator settings/parameters/controls
- Classification of Ventilators
- Complications of Positive Pressure Ventilation (PPV)

Introduction

- Mechanical ventilation is a medical process using a ventilator to help a patient breathe when they are unable to do so adequately on their own. The machine assists with moving air into and out of the lungs, delivering oxygen and removing carbon dioxide. It is used in cases of respiratory failure and can involve invasive methods like a breathing tube or noninvasive methods like a face mask.

Figure 11 Mechanical ventilator for positive pressure ventilation



Definition

Mechanical Ventilator is a **positive or negative** pressure breathing device that can maintain ventilation and oxygen delivery for a prolonged period(**Artificial lung**).



Indications of Mechanical Ventilation

- **Apnea : absence of breathing**(e.g., neuromuscular or cardiopulmonary collapse)
- **Acute ventilatory failure**, which is generally defined as a pH of less than or equal to 7.25 with (PaCO_2) greater than or equal to 50 mm Hg.
- **Severe hypoxemia:** (PaO_2) of less than or equal to 50 mm Hg on room air indicates a critical level of oxygen in the blood.
- **Respiratory muscle fatigue:** The muscles of respiration can become **fatigued** e.g., chronic lung disease are examples COPD.

1 Respiratory failure حالات فشل التنفس

في شوية علامات بتقولك العيان ده
محتاج يتحط علي جهاز الفنت

1 SaO₂

نسبة تشبع
الهيموجلوبين بالاكسجين

↓ 60%
Normal Range
(95-100%)

PaO₂ 2

الأكسجين الموجودة
في الدم الشرياني

↓ 50 mmHg
Normal Range
(80-100)

3 PaCo₂

↑ 50 mmHg
Normal Range
(35-45 mmHg)

pH 4

↓ 7.25
Normal Range
(7.35-7.45)

5 tachypnea

زيادة في معدل التنفس
↑ 30 c/m

bradypnea 6

انخفاض في معدل التنفس

ملحوظة
↓ 80 = Mild hypoxemia
↓ 70 = Moderate hypoxemia
↓ 60 = Sever hypoxemia

Indications of Mechanical Ventilation



Common ventilator settings/parameters/controls

Tidal Volume (V_t): Is the volume of gas delivered to a patient during a ventilator breath? i.e. the amount of air inspired and expired with each breath.

(Usual volume selected is between 5 to 8 ml/kg body weight)

Respiratory Rate (f): Is the number of breaths the ventilator will deliver/minute (12-20 b/m).

Minute Volume(MV): Is the volume of the air in one minute ($V_T \times f$)

Common ventilator settings/parameters/controls

Positive end expiratory pressure (PEEP) Positive pressure applied at the end of expiration of ventilator breaths expiration to maintain alveolar recruitment during mandatory breath. Usual setting 5: 10cm H₂O

Pressure support (PS) preset pressure delivered by ventilator to assist spontaneous breath. Usual setting: 10-15cm H₂O

Common ventilator settings/parameters/controls

Fraction of inspired oxygen (FIO₂): The percent of oxygen concentration that the patient is receiving from the ventilator. (Between 21% & 100%)usual amount 40-60%.

I:E Ratio (inspiration to expiration ratio): Is the ratio of inspiratory time to expiratory to expiratory time during breath (Usually (1:2)



Oxygen
الأكسجين

Air
الهواء



expiration
الهواء الذي خارج من العيان

inspiration
الهواء الذي داخل للعيان

Humidifier
المرطب



Common ventilator settings/parameters/controls

Humidification and Airway temperature

Humidifier temperatures should be kept close to body temperature 35°C-37°C.

Decreased humidity will cause dried secretions and plugging.

The humidifier should be checked for adequate water levels.

Humidifier should not be overfilled as this may increase circuit resistance and interfere with spontaneous breathing..

Classification of Ventilators

Ventilators are categorized as either

- **Negative pressure**
- **Positive pressure.**

Negative Pressure Ventilators

- Negative-pressure ventilators exert a negative pressure on the external chest.
 - Decreasing the intrathoracic pressure during inspiration allows air to flow into the lung.
- Physiologically, this type of assisted ventilation is similar to spontaneous ventilation.



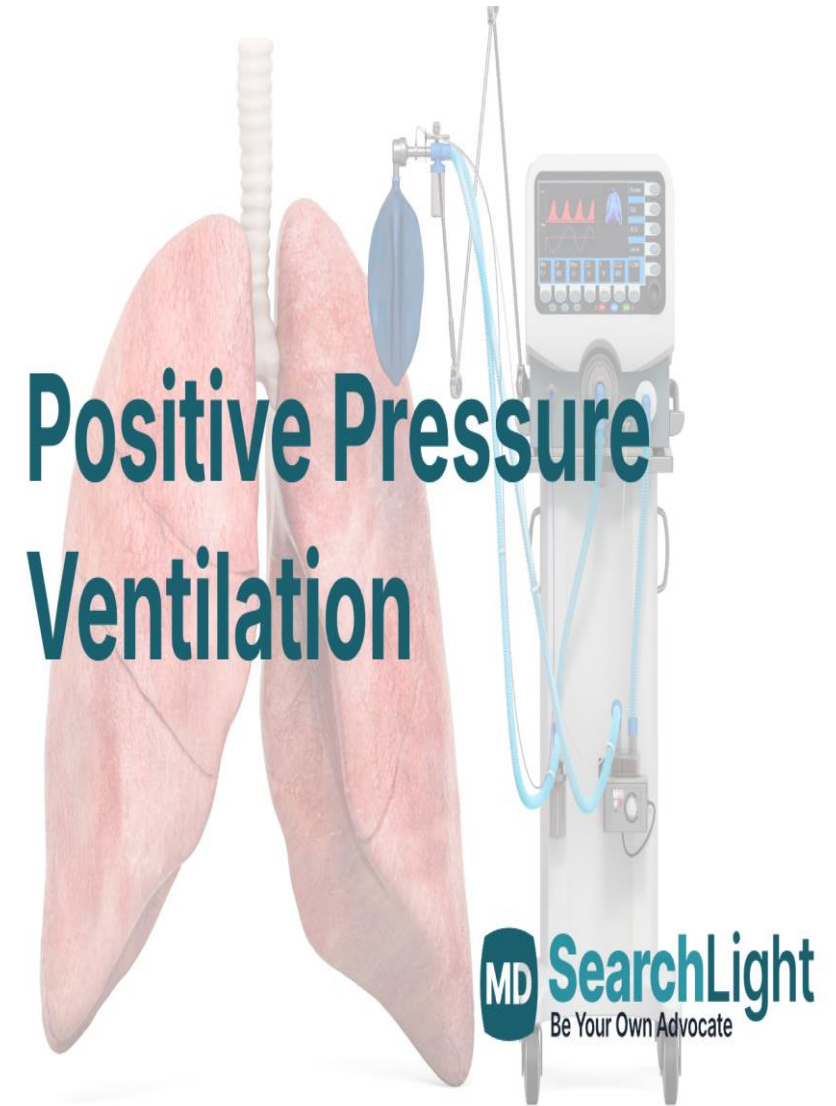
Negative Pressure Ventilators

- It is used in chronic respiratory failure associated with neuromuscular conditions, such as poliomyelitis, muscular dystrophy, and myasthenia gravis.
- Negative- pressure ventilators do not require intubation of the airway.
- There are several types of negative pressure ventilators: iron lungs, and chest cuirass.



Positive Pressure Ventilation

- Positive-pressure ventilators inflate the lungs by exerting positive pressure on the airway, pushing air in, forcing the alveoli to expand during inspiration. Expiration occurs passively.



Positive Pressure Ventilation

1- Invasive Mechanical Ventilation:

Endotracheal intubation or tracheostomy is usually necessary for Positive-pressure ventilators.

2- Noninvasive positive-pressure ventilation does not require intubation.

Invasive

اختراقي

Non-invasive

غير اختراقي

Endotracheal tube



الانبوبة الحنجرية

Tracheostomy



الشق الحنجري

Face mask



EgyptNursology

إيجي نيرسولوجي

×

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🔊

20:04

11:55

⏸

Positive-pressure modes of ventilation

have traditionally been categorized into

- **Volume ventilation**
- **Pressure ventilation**

•

Volume Ventilation

- Volume ventilation has traditionally been the most popular form of PPV, largely because **Tidal Volume (V_t) and Minute Ventilation (MV)** are ensured, which is an essential goal in the patient with acute illness.
- The preset volume of air delivered with each inspiration. Once this preset volume is delivered to the patient, the ventilator cycles off and exhalation occurs passively.
- A preset V_t is delivered with each breath regardless of resistance and compliance. Tidal Volume is stable from breath to breath, but airway pressure may vary.

Volume Modes

1. *Control Ventilation (CV) or Controlled Mandatory Ventilation (CMV)*
2. *Assist/Control (A/C) or Assisted Mandatory Ventilation (AMV)*
3. *Synchronized Intermittent Mandatory Ventilation (SIMV)*

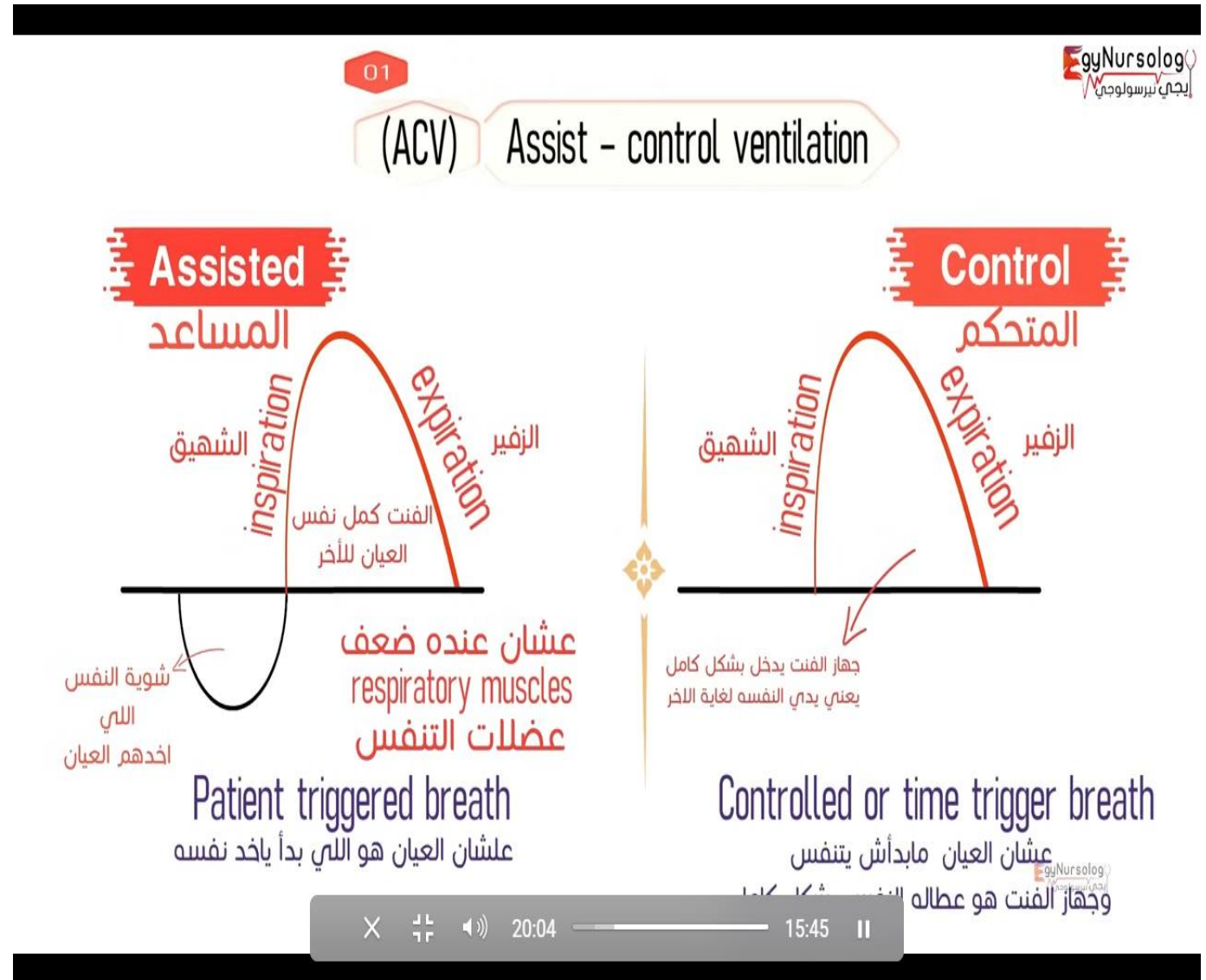
Volume Modes

- *Control Ventilation (CV) or Controlled Mandatory Ventilation (CMV)*

With this mode, the ventilator provides all the patient's minute ventilation. The clinician sets the **respiratory rate, V_t , inspiratory time, and positive End Expiratory Pressure (PEEP).**

Volume Modes

- *Assist/Control (A/C) or Assisted Mandatory Ventilation (AMV)* When the patient initiates a spontaneous breath. The ventilator sensitivity also is set, and a full-volume breath is delivered. This option requires that a rate, V_t , inspiratory time, and PEEP be set for the patient.



Volume Modes

○ *Synchronized Intermittent Mandatory Ventilation (SIMV)*

This mode requires that rate, V_t , inspiratory time, sensitivity, and PEEP are set by the clinician. In between mandatory breaths, patients can spontaneously breathe at their own rates and V_t . With SIMV, the ventilator synchronizes the mandatory breaths with the patient's own inspirations.

Synchronized Intermittent Mandatory Ventilation (SIMV)

EgyNursology | شرح جهاز التنفس الصناعي (MV) - الجزء الأول

يخالي العيان يأخذ نفسه طبيعي سواء اخذه قصير او طويل بمعدل بطي او سريع فهو هيسب الرئة تشتغل ومشي هيدخل نهاني لكن لو العيان وقف النفس

Synchronized IMV

Spontaneous Breaths

Ventilator Breath

ضعف في respiratory muscles

علطول جهاز الفت هيدخل بشكل كامل وهيغطي له النفس كله

MORE VIDEOS

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YouTube

Pressure Ventilation

- The clinician selects the desired pressure level and the V_t is determined by the selected pressure level, resistance, and compliance.

Types Pressure mode

- 1. Pressure Support Ventilation (PSV)*
- 2. Pressure-Controlled/Inverse Ratio Ventilation (PC/IRV)*
- 3. Non-invasive Bilateral Positive-Pressure Ventilation Mode(BiPAP)**
- 4. Continuous Positive Airway Pressure (CPAP)*

Pressure Modes

1- Pressure-Controlled

- Which delivers a preset pressure for a set time

2- Pressure Support Ventilation (PSV)

- Patient **initiates** each breath
- Ventilator **assists** with a fixed pressure level
- Patient controls **rate, timing, and tidal volume**

It's commonly used for:

- Weaning from mechanical ventilation
- Providing comfort while maintaining some ventilatory assistance

Non-invasive Bilateral Positive-Pressure Ventilation Mode

- is a noninvasive form of mechanical ventilation provided by means of a nasal mask, nasal prongs, or a full face mask.
- It is used in the treatment of patients with chronic respiratory insufficiency to manage acute or chronic respiratory failure without intubation.
- The system allows the clinician to select two levels of positive-pressure support: an inspiratory pressure support level (referred to as IPAP) and an expiratory pressure called EPAP (PEEP/CPAP level).



1

Bi PAP
(NIV)

Non-Invasive Ventilation

ودا علشان مش بيتوصل إلا بطريقة
Non-Invasive

2

Inspiratory pressure
لوحد

هو يتحكم في

Expiratory pressure
لوحد

Inspiratory positive airway pressure

IPAP

8-20 cm H₂O

High pressure

Expiratory positive airway pressure

EPAP

5-8 cm H₂O

Low pressure

- **BiPAP** is beneficial in patients with worsening hypoventilation, obstructive apneic episodes, or both.
- It is also useful to prevent intubation in patients with respiratory failure and hypercarbia as well as to prevent reintubation following extubation in borderline cases.
- Use of a full face mask may increase the risk for aspiration and risk for re-breathing carbon dioxide; therefore, ventilation with a full face mask should be used cautiously.
- Thick or profuse secretions and poor cough may be relative contraindications for BiPAP.



Continuous Positive Airway Pressure (CPAP)

- Continuous positive airway pressure (CPAP) provides positive pressure to the airways throughout the respiratory cycle.
- CPAP assists spontaneously breathing patients to improve their oxygenation by elevating the end-expiratory pressure in the lungs keeping the upper airway and trachea open throughout the respiratory cycle.
- CPAP can be used for intubated and non-intubated patients.

To use CPAP, the patient must be breathing independently.



Ventilator Alarms

Pressure Alarms

- *High-pressure alarms* : causes for alarm activation include **secretions**, condensate in the tubing, biting on the endotracheal tubing, increased **resistance** (i.e., bronchospasm), decreased compliance (e.g., pulmonary edema, pneumothorax), and tubing compression.
- *Low-pressure alarms* are used to sense disconnection, circuit leaks, and changing compliance and resistance.



High pressure

هنا الجهاز بيدي الهواء ويلاقى الطريق مقفول

بتبص علي كل وصلات الفنت
وتشوف المشكلة فين وتحلها
بانك تعدل الوصلة المتنية او
تغير الوصلة كلها

مممكن تكون وصلة من
وصلات الفنت متنية



Endotracheal tube



ازاي تتيب العيان صح؟



↑ secretions

اللي موجودة في الافرازات في مجري
مجري التنفس والتبوة التنفس والتبوة

suction

علشان تسحب كل
secretions

مممكن تلاقي المريض بيعض ع التبوة
oral airway



في فم العيان وتثبتها كويس
عشان متتحركش من مكانها
او تديله جرعة الدورميكس اللي
المفروض بياخذها



3

Low pressure

الضغط قليل والجهاز بيدي الهواء ف الفراغ



تسريب من



تسريب في

اننا بنثبت به التيوبة وبالتالي
بيمنع تسريب الهواء علشان
التيوبة مثبتة ف مكانها

Cuff

في الحالة اللي زي
دي انت بتغير التيوبة

Low pressure

معناه اي تسريب من اي حته



ثقب في اي وصلة من وصلات الفنت
المفروض انك تغير الوصلة

اي وصلة من وصلات الفنت حصلها
disconnection

يعني اتفكت او ممكن تكون
وصلة الفنت اللي متوصلة

Endotracheal tube

هي اللي اتفكت

المفروض تعمل شيك لكل

الوصلات وتتأكد انهم متوصلين كويس



4 High respiratory rate زيادة في معدل التنفس

Anxiety
العيان قلقان

1

2

Pain
العيان بيتألم

3

Hypoxia
انخفاض في الاكسجين

4

Fever
ارتفاع في درجة الحرارة

هعالج الحاجات دي ازاوي؟
order
لان العلاج بيختلف من حالة لحالة



Alarm silence or pause

- Options are built in by ventilator manufacturers so that clinicians can temporarily silence alarms for short periods (i.e., 20 seconds) because alarms must stay activated at all times.

Complications of Positive Pressure Ventilation (PPV)

1. Ventilator induced lung Injury

Ventilator can cause two types of lung injury:

- A) *Air leaks*
- B) *Biotrauma*

Ventilator induced lung Injury

Air leaks related to ventilator are the result of

- ❑ **Volutrauma** (excessive volume in the alveoli)
- ❑ **Barotrauma** (excessive pressure in the alveoli)
- ❑ **Atelectrauma** (Shearing of the alveoli due to repeated opening and closing of the alveoli)

- **Volutrauma, barotrauma, and atelectrauma** can lead to excessive alveolar wall stress and damage to the alveolar –capillary membrane, resulting in air leakage into surrounding spaces.
- The air then travels out through the hilum and into the mediastinum (pneumomediastinum), pleural space (**pneumothorax**), pericardium (pneumopericardium), peritoneum (pneumoperitoneum), and subcutaneous tissue (subcutaneous emphysema).

Volutrauma, barotrauma, and atelectrauma can cause the release of cellular mediators and initiation of inflammatory- immune response which is known as

Biotrauma.

- Biotrauma can result in to the development of ARDS.
- ARDS is a life-threatening condition that occurs when there is severe fluid buildup in both lungs. The fluid buildup prevents the lungs from working properly which inhibit the transfer of oxygen from air into the body and carbon dioxide out of the body into the air

To limit **ventilator induced lung injury**

- Control peak and plateau pressures
- Optimizing PEEP, Vt should be set at $< 10 \text{ Ml /Kg}$
- Prevent auto-PEEP
- Ensure patient-ventilator synchrony
- Ensure proper artificial airway position.

Hemodynamic changes

- PPV can **reduce venous return**, due to increase intrathoracic pressures which are transmitted to the adjacent capillaries.

Gastrointestinal Problems

- Gastrointestinal complications associated with mechanical ventilation include distention (from air swallowing), hypomotility and ileus (from immobility and the use of narcotic analgesics), and vomiting.
- Maintenance of an adequate bowel elimination pattern is necessary to prevent abdominal distention with resulting impingement on diaphragmatic excursion.

Complications associated with Immobility

- Muscle wasting and weakness
- Contractures
- Loss of skin integrity
- Deep venous thrombosis that can result in pulmonary embolus
- Constipation.

Aspiration

Aspiration can occur before, during, or after intubation.

The risk for aspiration after intubation can be **minimized by**

- ❖ Maintaining appropriate cuff inflation
- ❖ Evacuating gastric contents
- ❖ Relieving distention with suction
- ❖ Suctioning the oropharynx (especially before cuff deflations),
- ❖ Elevating the head of the patient's bed 30 degrees or more at all times.

Ventilator-Associated Pneumonia (VAP)

➤ Ventilator-associated pneumonia (VAP) is a subgroup of hospital-acquired pneumonia that refers to the development of pneumonia 48 to 72 hours after endotracheal intubation .

A great risk for development of pneumonia exists after placement of an artificial airway because the tube bypasses or impairs many of the lungs normal defense mechanisms

Ventilator-Associated Pneumonia (VAP)

- The use of respiratory therapy devices (ventilators, nebulizers, intermittent positive-pressure breathing machines) also can increase the risk of pneumonia .
- The severity of the patient's illness and the presence of **ARDS or malnutrition** significantly increase risk for VAP .
- **Therapeutic measures such as**
 - Nasogastric tubes promote aspiration by acting as a wick for stomach contents, whereas enteral feedings, antacids, histamine inhibitors, and proton-pump inhibitors increase the pH level of the stomach, promoting the growth of bacteria that can then be aspirated

Ventilator-Associated Pneumonia (VAP)

Prevention of VAP is critical. The Institute for Healthcare Improvement (IHI) identified five interventions Known as the “IHI **Ventilator Bundle**” these interventions are

- Elevation of the head of the bed (30 to 45) degrees
- Daily “sedation vacations” and assessment of readiness to extubate.
- Peptic ulcer disease prophylaxis
- Deep venous thrombosis prophylaxis
- Daily oral care with chlorhexidine (every 12 hours)

Weaning From Mechanical Ventilation

Definition

Weaning from mechanical ventilation is the gradual withdrawal of the mechanical ventilation and reestablishment of spontaneous breathing.

The process to achieve this goal includes

- ☐ Correcting the cause of respiratory failure
- ☐ Preventing complications,
- ☐ Restoring or maintaining physiological and psychological functional status.

- * **Short-term mechanical ventilation** (3 days or less)
- * **Long-term mechanical ventilation** (more than 3 days).
- * Each patient is evaluated daily for readiness to wean by performing a spontaneous breathing trial.

Weaning from Short-term mechanical ventilation

Readiness Criteria

- ❖ Hemodynamically stable, and not requiring vasoactive support drugs.
- ❖ SaO₂ greater than 90% on FiO₂ 40% or less, PEEP 5 cm H₂O or less
- ❖ Chest radiograph reviewed for correctable factors; treated as indicated
- ❖ Metabolic indicators (serum pH, major electrolytes) within normal range

Readiness Criteria

- . Hematocrit more than 25%
- . Core temperature more than 36°C and less than 39°C
- . Adequate management of pain/anxiety/agitation
- . No residual neuromuscular blockade agent
- . ABGs values normalized or at patient's baseline

Tolerance Criteria

If the patient displays any of the following, the weaning trial should be stopped and the patient returned to “rest” settings.

- Sustained respiratory rate greater than 35 breaths/min
- SaO₂ less than 90%
- Tidal Volume 5 mL/kg or less
- Sustained minute ventilation greater than 200 mL/kg/min

Tolerance Criteria

- Evidence of respiratory or hemodynamic distress, which include:
 - Labored respiratory pattern (using accessory muscles, Sternocleidomastoid)
 - Increased anxiety, diaphoresis, or both o Sustained heart rate greater than 20% higher or lower than baseline
 - Systolic blood pressure exceeding 180 mm Hg or less than 90 mm Hg

Methods of Weaning from Mechanical Ventilator

- ❖ T- piece trial
- ❖ Synchronized Intermittent Mandatory Ventilation (SIMV) Mode
- ❖ Continuous Positive Airway Pressure (CPAP)
- ❖ Pressure-Support Ventilation Mode



Thank
you

